Layers

IT5100D AY25/26 S2

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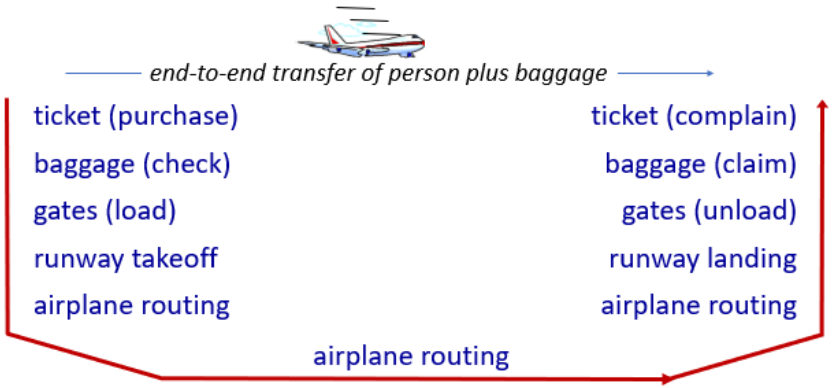
Layers

- Networks are complex, with many “pieces”
  - Hosts, routers, applications, protocols, …
- **Layers:** Approach to *designing / discussing* complex systems

Air Travel Analogy

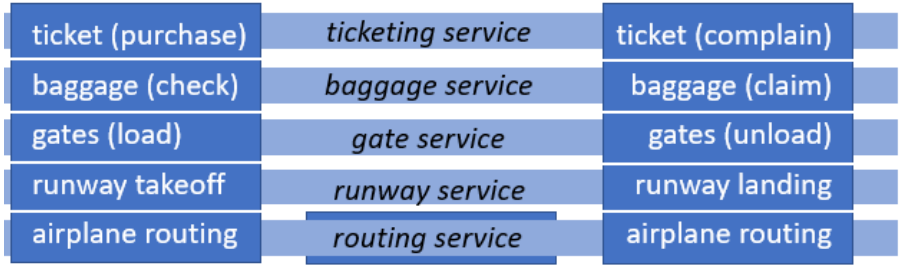
The *system* of air travel:

a series of steps involving many services



Each layer implements a service

- Via its own *internal layer actions*
- *Relying on* services provided by layer below



Layers

- Explicit structure allows identification, relationship of system's pieces
  - Layered *reference model* for discussion
- *Modularization* eases maintenance, updating of system
  - Change in a layer's service implementation is transparent to rest of system
  - E.g., changes in airport gate procedure does not affect the rest of the system

Layered Internet Protocol Stack

- **Application:** supporting network applications
  - HTTP, IMAP, SMTP, DNS
- **Transport:** process-process data transfer
  - TCP, UDP
- **Network:** routing of datagrams from source to destination
  - IP, routing protocols
- **Link:** data transfer between neighbouring network elements
  - Ethernet, 802.11 (WiFi), PPP
- **Physical:** bits “on the wire”

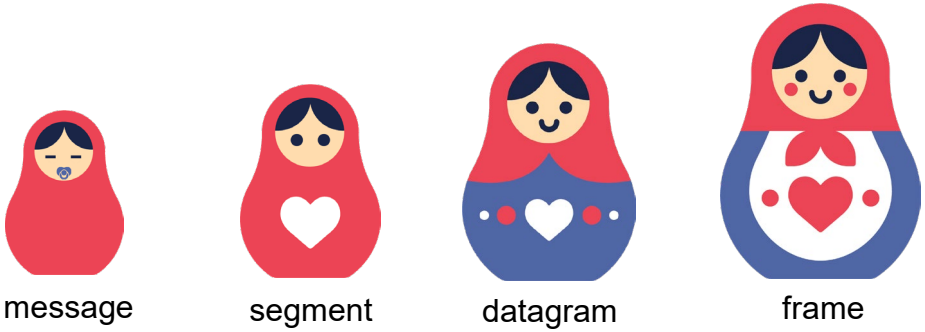
Protocol Layer Communication

- Within a host, layers talk only to the layer *just below or above*.
  - Each layer does not need to care how any of the other layers are implemented – they use *inter-layer APIs*.
- Each layer talks to the *corresponding layer on the other host* via some sort of protocol.

Encapsulation

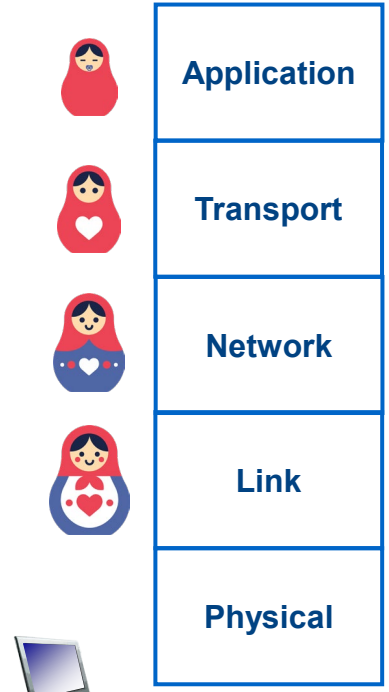
- The protocol at each layer encapsulates the message from the layer above it with a header
- The header contains information used to implement that layer's service

Matryoshka dolls (stacking dolls)



Credit: https://dribbble.com/shots/7182188-Babushka-Boi

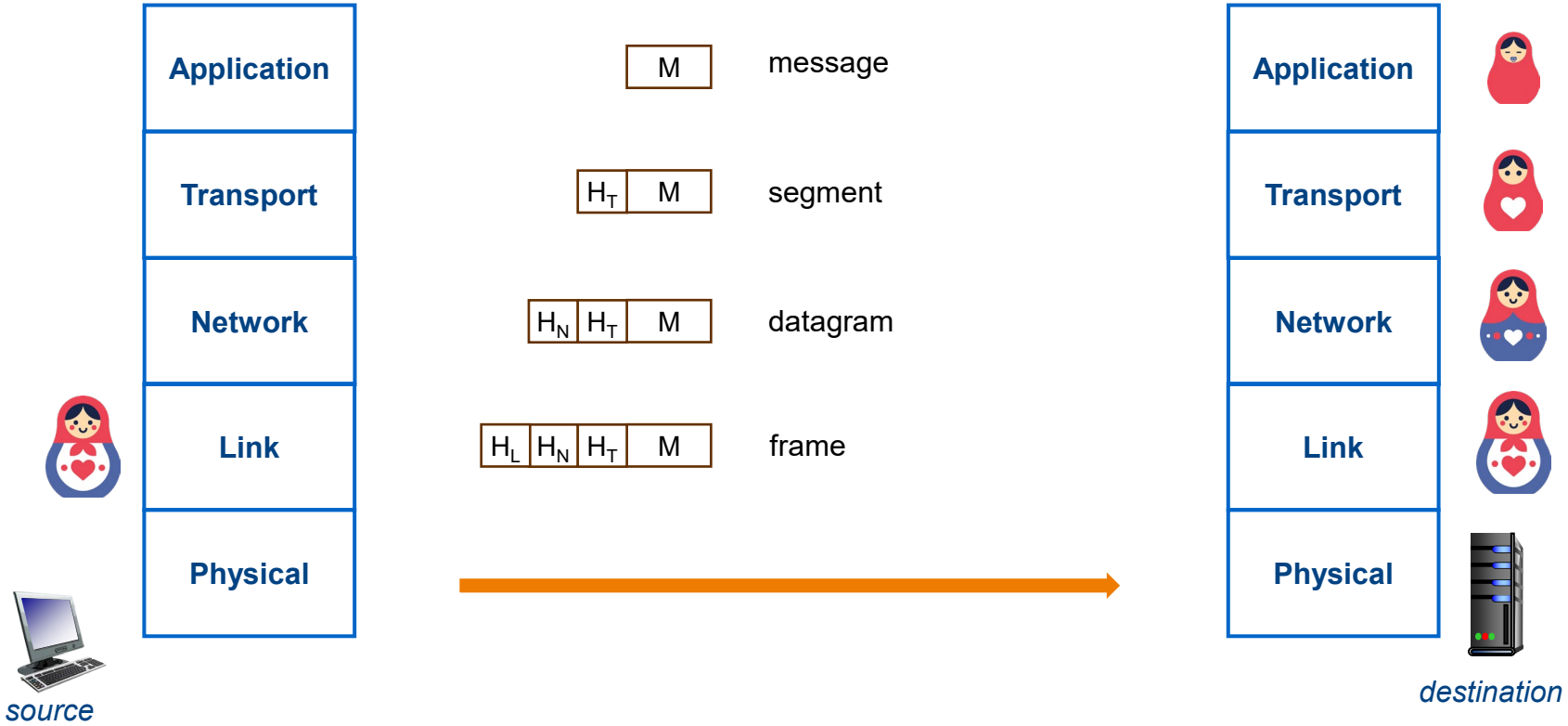
Services, Layering and Encapsulation



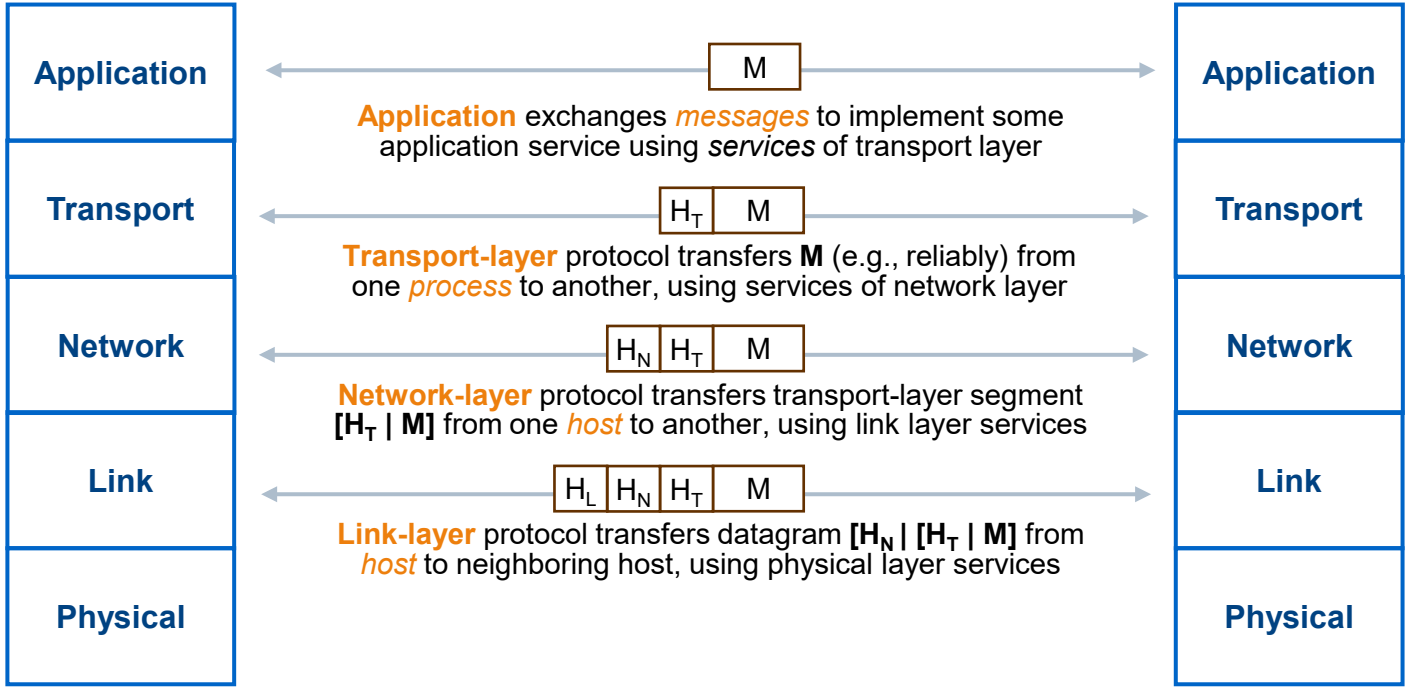
source

M	message
H_T M	segment
H_N H_T M	datagram
H_L H_N H_T M	frame

Services, Layering and Encapsulation



Services, Layering and Encapsulation



Application exchanges *messages* to implement some application service using *services* of transport layer

Transport-layer protocol transfers **M** (e.g., reliably) from one *process* to another, using services of network layer

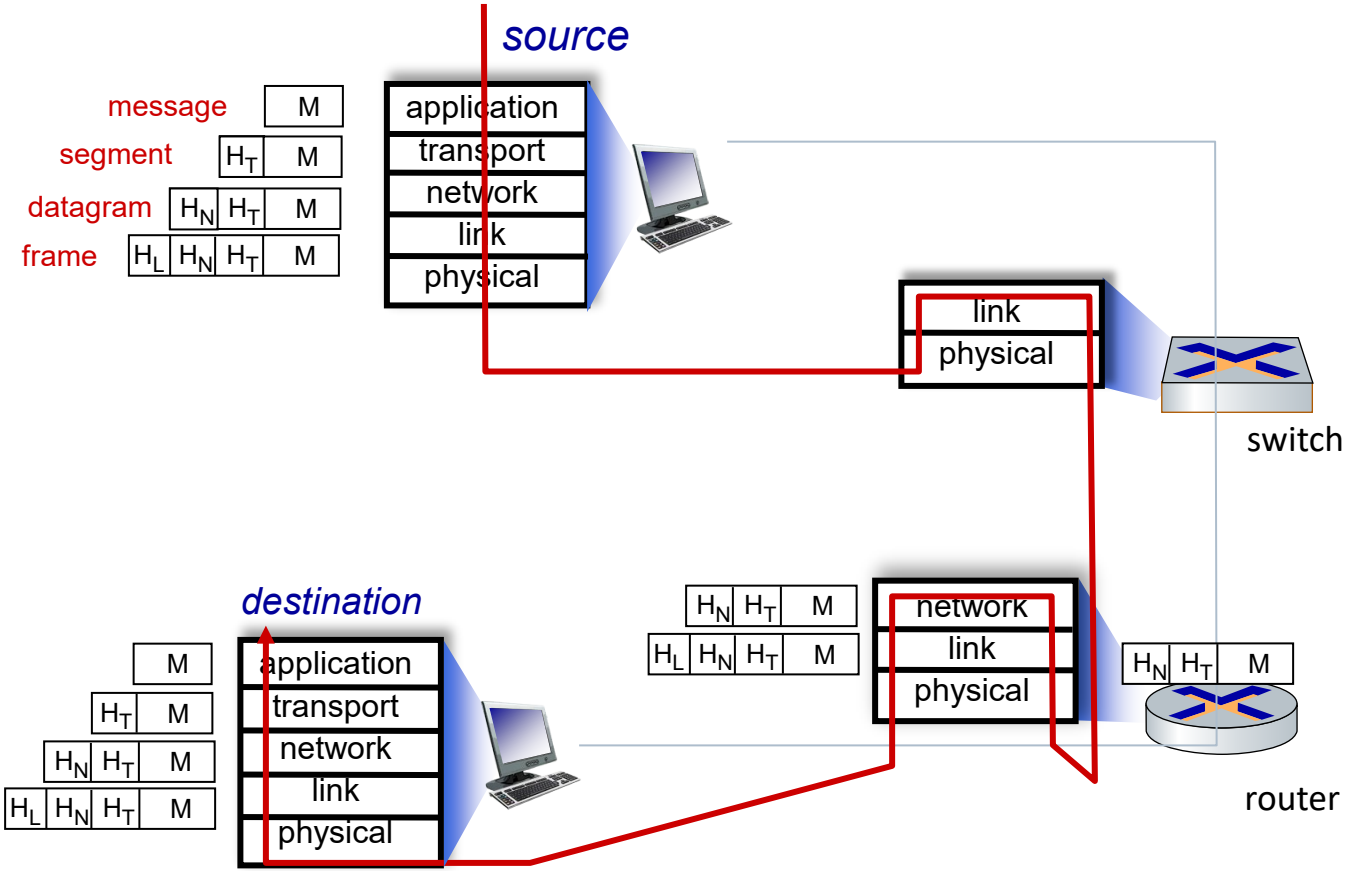
Network-layer protocol transfers transport-layer segment **[H_T | M]** from one *host* to another, using link layer services

Link-layer protocol transfers datagram **[H_N | [H_T | M]** from *host* to neighboring host, using physical layer services

source

destination

Encapsulation: An End-to-End View



See on Wireshark

*WiFi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
151	13.796321	192.168.0.147	45.60.33.225	ICMP	74	Echo (ping) request id=0x0001, seq=74/18944, ttl=128 (reply in 152)
152	14.073125	45.60.33.225	192.168.0.147	ICMP	74	Echo (ping) reply id=0x0001, seq=74/18944, ttl=52 (request in 151)
203	14.812371	192.168.0.147	45.60.33.225	ICMP	74	Echo (ping) request id=0x0001, seq=75/19200, ttl=128 (reply in 210)
210	15.077703	45.60.33.225	192.168.0.147	ICMP	74	Echo (ping) reply id=0x0001, seq=75/19200, ttl=52 (request in 203)
232	15.828244	192.168.0.147	45.60.33.225	ICMP	74	Echo (ping) request id=0x0001, seq=76/19456, ttl=128 (reply in 235)
235	16.094011	45.60.33.225	192.168.0.147	ICMP	74	Echo (ping) reply id=0x0001, seq=76/19456, ttl=52 (request in 232)
250	16.834288	192.168.0.147	45.60.33.225	ICMP	74	Echo (ping) request id=0x0001, seq=77/19712, ttl=128 (reply in 255)
255	17.102611	45.60.33.225	192.168.0.147	ICMP	74	Echo (ping) reply id=0x0001, seq=77/19712, ttl=52 (request in 250)

```
> Frame 152: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{842EEA67-1F87-49F8-8718
> Ethernet II, Src: DLinkInterna_f7:2a:c4 (e4:6f:13:f7:2a:c4), Dst: Intel_80:e6:43 (30:05:05:80:e6:43)
v Internet Protocol Version 4, Src: 45.60.33.225, Dst: 192.168.0.147
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x28 (DSCP: AF11, ECN: Not-ECT)
    Total Length: 60
    Identification: 0x2948 (10568)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 52
    Protocol: ICMP (1)
    Header Checksum: 0x4cf9 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 45.60.33.225
    Destination Address: 192.168.0.147
v Internet Control Message Protocol
    Type: 0 (Echo (ping) reply)
    Code: 0
    Checksum: 0x5511 [correct]
    [Checksum Status: Good]
    Identifier (BE): 1 (0x0001)
    Identifier (LE): 256 (0x0100)
    Sequence Number (BE): 74 (0x004a)
    Sequence Number (LE): 18944 (0x4a00)
    [Request frame: 151]
    [Response time: 276.804 ms]
  v Data (32 bytes)
      Data: 6162636465666768696a6b6c6d6e6f707172737475767761626364656667686 9
      [Length: 32]
```

```
0000  30 05 05 80 e6 43 e4 6f  13 f7 2a c4 08 00 45 28   0····C·o ··*···E(
0010  00 3c 29 48 00 00 34 01  4c f9 2d 3c 21 e1 c0 a8   ·<)H··4· L·-<!···
0020  00 93 00 00 55 11 00 01  00 4a 61 62 63 64 65 66   ····U··· ·Jabcdef
0030  67 68 69 6a 6b 6c 6d 6e  6f 70 71 72 73 74 75 76   ghijklmn opqrstuv
0040  77 61 62 63 64 65 66 67  68 69                     wabcdefg hi
```

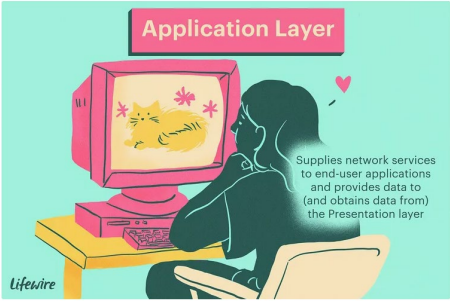
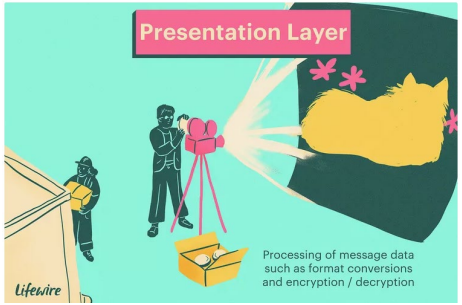
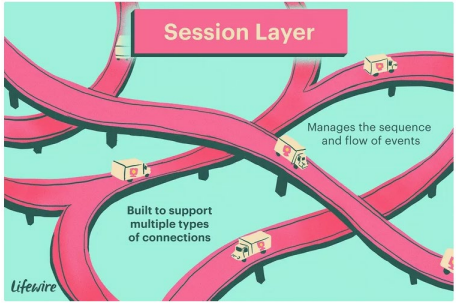
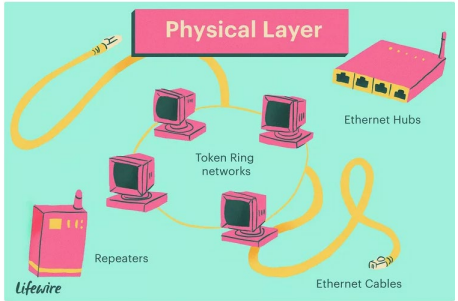
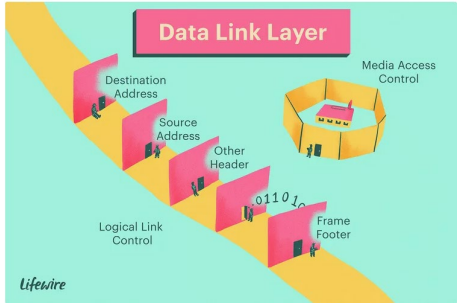
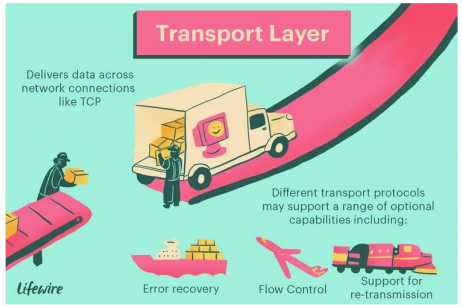
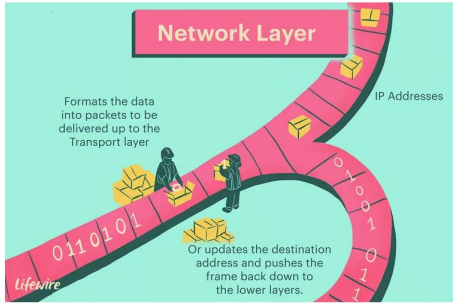
Internet Applications

- All Internet applications work over the same transport layers.
- Applications just need to know:
  1. *IP address* of the other party (what host the other party is running on—a network layer concept), and
  2. *Port number* of the application running at the other end (because the other machine might be running multiple services—a transport layer concept)
- Applications pass these two pieces of information to the transport layer to make the communication happen.
- We will learn more about these in coming lectures!

Note on Other Reference Models

Arpanet Reference Model (RFC 871)	Internet Standard (RFC 1122)	Internet model (Cisco Academy[38])	TCP/IP 5-layer reference model (Kozierok,[39] Comer[40])	TCP/IP 5-layer reference model (Tanenbaum[41])	TCP/IP protocol suite or Five-layer Internet model (Forouzan,[42] Kurose[43])	TCP/IP model (Stallings[44])	OSI model (ISO/IEC 7498-1:1994[45])
Three layers	*Four layers*	*Four layers*	*Four+one layers*	*Five layers*	*Five layers*	*Five layers*	*Seven layers*
Application/ Process	Application	Application	Application	Application	Application	Application	Application
							Presentation
							Session
Host-to-host	Transport	Transport	Transport	Transport	Transport	Host-to-host or transport	Transport
	Internet	Internetwork	Internet	Internet	Network	Internet	Network
Network interface	Link	Network interface	Data link (Network interface)	Data link	Data link	Network access	Data link
—	—	—	(Hardware)	Physical	Physical	Physical	Physical

Source: https://en.wikipedia.org/wiki/Internet_protocol_suite

An Illustration

- OSI model illustration by Lifewire